

Self-Administered Tinnitus Pitch Matching versus a Conventional Audiometric Procedure

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Keywords

Tinnitus · Frequency · Computer-based method · Pitch matching

Abstract

Objective: Obtaining an accurate tinnitus pitch match is an initial and critical requirement for tinnitus evaluation and treatment, particularly for applying tailor-made notched music training. We investigated whether computer-based self-administered tinnitus pitch matching (CSTPM) is comparable with a conventional audiometric procedure (CAP).

Methods: In total, 82 patients (mean age 45.52 years; 42 females) with tonal tinnitus participated. The CAP was performed by the same audiologist using a 2-alternative forced choice method with a frequency range of 0.25–16 kHz. In the CSTPM, the subjects used personal computer software with a scrolling slider to select the sound closest to their tinnitus pitch. After each matching procedure, an octave challenge test was applied. A multivariate logistic regression was performed to determine factors associated with the difference between the CSTPM and CAP. **Results:** The subjects' mean hearing threshold was 21.25 ± 17.61 dB HL; the mean tinnitus handicap inventory score was 35.56 ± 24.09 . The mean pitches measured with the CSTPM and CAP were 6.29 ± 4.30 and 6.98 ± 5.33 kHz, respectively. In total, 57 (69.5%)

subjects matched their tinnitus with less than half an octave difference between the procedures. The results of the 2 methods correlated significantly with each other (Pearson $r = 0.793$, $p < 0.001$). Octave confusion was a significant factor affecting the difference between the procedures (odds ratio 8.92, $p < 0.05$). **Conclusions:** The CSTPM appears to be as accurate as the standard audiological procedure, and may be used instead of the CAP when octave confusion is minimized.

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Introduction

Tinnitus is the perception of sound in the absence of an external sound source. Tinnitus assessment and quantification is an important issue in providing effective counseling, establishing a reference point for diagnosis, categorizing and managing patients, finding a proper treatment, and developing new therapies. Tinnitus is a subjective sensation; there is currently no way to objectively measure tinnitus, and existing protocols for measurement of the patient's perception of tinnitus rely exclusively on patient feedback.

The psychoacoustic measures of tinnitus that are usually estimated during its evaluation are pitch, loudness,

maskability, and residual inhibition. Among them, pitch is the fundamental measure that attempts to quantify the tinnitus frequency. A matched tinnitus pitch can be used as a reference for the selection and fitting of sound therapies that have been gaining popularity and showing increasingly successful results. In particular, tailor-made notched music training was developed to suppress hyperactive-at-resting auditory cortical neurons, which represent the tinnitus frequency, by reinforcing lateral inhibition through listening to music with an octave range centered at the tinnitus frequency filtered out [Okamoto et al., 2010; Teismann et al., 2011; Pantev et al., 2012].

Thus, accurate tinnitus pitch matching is an important step for successful sound-based tinnitus treatment [Pantev et al., 2012; Tass et al., 2012]. Tinnitus pitch matching has traditionally been performed as part of audiological evaluations for tinnitus, typically requiring patients to visit hospitals or specialist facilities for audiometry. Various attempts have been made to automate tinnitus assessment. Computer- and mobile-based protocols have been proposed for automated tinnitus measurements [Henry et al., 2004; Mahboubi et al., 2012; Wunderlich et al., 2015]. Recently, several online and mobile applications for tailor-made notched music training have been introduced (e.g., AudioNotch <http://www.audionotch.com>, Tinnitus Pro <http://promedicalaudio.com>, Tinnitus Off <http://www.tinnitus-off.com>, and Tinnitus Buster <http://tinnitus.kangwon.ac.kr>), which offer self-conducted tinnitus pitch assessments. Subsequently, the sound files provided are modified automatically for further therapy, considering the determined tinnitus frequency. With the increasing popularity of such automated procedures, the validity and reliability of their self-administered tinnitus pitch-matching technique requires an in-depth evaluation. The purpose of this study was to evaluate the reliability of a computer-based self-administered tinnitus pitch matching (CSTPM), implemented as a part of tailored music therapy software Tinnitus Buster (<http://tinnitus.kangwon.ac.kr>), by comparing it with a conventional audiometric procedure (CAP). Factors contributing to the difference between the results from the 2 procedures were investigated.

Materials and Methods

In total, 82 patients with tonal tinnitus that was stable in pitch over time participated in this study. Patients with oncological, neurological, or psychiatric disorders were excluded. The study protocol was reviewed and approved by the Institutional Review Board of Kangwon National University Hospital. All patients were

informed of the purpose and procedures of the study and signed a statement of informed consent prior to participation.

Prior to the tinnitus pitch-matching procedures, pure tone hearing thresholds were measured by an audiologist in the frequency range of 250 Hz to 16 kHz in a double-walled soundproof room (ISO 8253-1) with an audiometer (MADSEN Astera, OTO-suite module; GN Otometrics, Taastrup, Denmark) and calibrated headphones (TDH39; GN Otometrics).

CAP

The CAP was performed by the same audiologist in the soundproof room using the same audiometric equipment. Pure tones were used for pitch matching. The testing was applied to the ear contralateral to the predominant or louder tinnitus, because this is believed to be less challenging than matching at the ipsilateral ear [Goodwin and Johnson, 1980; Henry et al., 1999]. If the tinnitus was equally loud on both sides, or was localized in the head, the testing was done in the ear with better hearing. The testing ear was chosen randomly if there was no difference between the tinnitus level/hearing acuity of the 2 ears. The procedure was a 2-alternative forced choice (2AFC). The patients were presented with 2 frequencies and were asked to select the frequency most resembling their tinnitus. The frequency range was 250 Hz to 16 kHz. The first presented tone pair was 250 Hz/4 kHz. The next presented tone pair would consist of the tone that the patient selected and a tone an octave higher or lower. For example, if the patient selected 4 kHz in the first pair, a 4 kHz/8 kHz pair was presented next. If the patient selected 250 Hz in the first pair, 125 Hz/250 Hz would be presented next. The procedure was further repeated in half-octave steps until the final frequency was chosen. The CAP procedure was repeated 3 times and, among the 3 resulting frequencies, the final frequency was selected by a 2AFC test starting with the lowest 2 frequencies until only 1 frequency remained. Finally, an octave confusion test was performed using a 2AFC test that compared the final selected tinnitus pitch frequency with frequencies an octave higher and lower. The octave confusion test was performed twice; the positive octave confusion result was accepted in the case when the patient's answers were inconsistent between the 2 tests.

CSTPM

The CSTPM was performed using a desktop computer (Samsung ATIV One 7; Samsung Electronics Co. Ltd, Hwaseong, Republic of Korea) with an Intel HDMI Audio (ALC269) 6.16.0.3154 sound card (Intel Corp., Ho Chi Minh City, Vietnam) and Realtek High Definition Audio 6.0.1.7090 sound driver (Realtek Semiconductor Corp., Hsinchu, Taiwan) in an office room with active noise-cancelling headphones (QuietComfort3; Bose, Framingham, MA, USA). The patients set the computer sound volume with a volume slider, using the frequency determined during the CAP. Then, a scrolling slider with a starting frequency of 1 kHz was used to select the tone closest to the tinnitus pitch in 100-Hz steps. The slider frequency range was 250 Hz to 16 kHz. The tones were sine waves generated by a program developed in-house, running in the MATLAB environment (v.8.6, MathWorks Inc., Natick, MA, USA) and using the formula:

$$y(n) = A \sin\left(2\pi \frac{f}{F_s} n\right),$$

where A is the amplitude of the sound, f is the frequency of the sound, and F_s is the sampling frequency (44.1 kHz in this study). The CSTPM procedure was repeated 3 times and a 2AFC test was

Table 1. Demographic data and tinnitus characteristics of the subjects

Measure	
Male:female	40 (48.8):42 (52.2)
Age, years	45.52±16.17
Pure tone average, dB HL	21.25±17.61
Tinnitus duration, months	35.23±43.45
THI score	35.56±24.09
Tinnitus-matched loudness, dB SL	5.67±6.70
Tinnitus lateralization (bilateral:right:left)	13 (15.9):23 (28.0):46 (56.1)
Tinnitus matching site (right:left)	28 (34.1):54 (65.9)
Octave confusion positive	12 (14.6)

Data are expressed as the prevalence (%) for categorical variables and as means ± SD for continuous variables. THI, tinnitus handicap inventory.

applied to determine the final frequency in the same manner as during the CAP.

Statistical Analysis

Before the analysis, all frequency values were converted to a logarithmic scale to calculate means and standard deviations. The back-transformed mean frequency data were used for data reporting and visualization. Statistical comparisons of the CAP and CSTPM were made using the Wilcoxon signed-rank test. The relationship between the results of the 2 procedures was examined using Pearson correlation analysis. All patients were divided into 2 groups according to the test result difference between the CSTPM and CAP: group A included patients whose tinnitus pitch difference was ≤0.5 octaves and group B included patients with a >0.5-octave difference. A multivariate logistic regression was performed to determine significant factors associated with test result differences between CSTPM and CAP. The odds ratio (OR) and 95% confidence interval for the test result difference were calculated. All analyses were performed using SPSS software (v.18.0 for Windows; SPSS Inc., Chicago, IL, USA). Differences were considered statistically significant at $p < 0.05$.

Results

Of the 82 patients, 40 (48.8%) were male and 42 (52.2%) were female. Their mean age was 45.52 ± 16.17 years. The mean hearing threshold was 21.25 ± 17.61 (dB HL). The mean tinnitus duration and mean tinnitus handicap inventory scores were 35.23 ± 43.45 months and 35.56 ± 24.09 , respectively. The mean matched tinnitus loudness was 5.67 ± 6.70 (dB SL). In total, 23 (28.0%) patients had unilateral right-sided or right-dominant tinnitus, 46 (56.1%) had unilateral left-sided or left-dominant tinnitus, and 13 (15.9%) had bilateral tinnitus. Twenty-eight (34.1%) patients matched tinnitus pitch in the right ear and 54 (65.9%) patients matched in the left

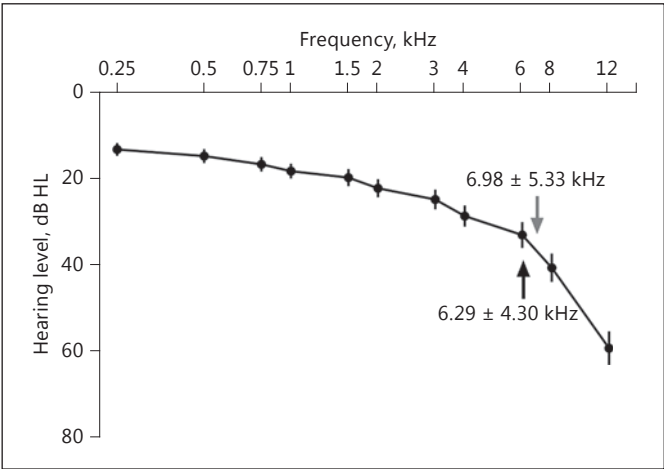


Fig. 1. Mean hearing thresholds (dB HL) for the 82 patients. Error bars represent standard errors. The black arrow shows tinnitus pitch measured with CSTPM, and the gray arrow shows tinnitus pitch measured with a CAP. $p = 0.369$ between CAP and CSTPM, tested using the Wilcoxon signed-rank test.

ear. Twelve (14.6%) patients experienced octave confusion (Table 1).

Figure 1 shows the mean hearing thresholds at each frequency for the 82 patients. The mean pitch-matched frequencies measured with the CSTPM and CAP were 6.29 ± 4.30 and 6.98 ± 5.33 kHz, respectively (Fig. 1). No statistically significant difference was found between the matched pitch frequencies determined by CAP and CSTPM ($p = 0.369$). A significant positive correlation between the results of the 2 procedures was observed ($r = 0.793$, $p < 0.001$; Fig. 2). The distribution of the matched pitches per each frequency was similar between the 2 methods (Fig. 3).

Table 2. Multiple logistic regression analysis results

	Group A (diff. ≤ 0.5 octave)	Group B (diff. > 0.5 octave)	OR	95% CI	<i>p</i> value
<i>n</i>	57 (69.5)	25 (30.5)			
Age, years	45.26 $\pm$ 16.43	46.12 $\pm$ 15.21	1.001	0.955–1.049	0.965
Male:female	27:30	13:12	2.165	0.615–7.620	0.229
Site (right:left)	21:36	9:16	0.807	0.213–3.050	0.752
PTA, dB HL	21.34 $\pm$ 16.01	20.1 $\pm$ 13.60	0.933	0.757–1.150	0.516
Low PTA, dB HL	15.70 $\pm$ 12.19	14.93 $\pm$ 14.42	1.075	0.933–1.237	0.316
High PTA, dB HL	30.87 $\pm$ 22.70	29.60 $\pm$ 17.10	1.007	0.911–1.112	0.895
THI	35.33 $\pm$ 23.82	36.08 $\pm$ 24.21	0.991	0.960–1.022	0.557
Loudness, dB SL	4.56 $\pm$ 4.61	8.2 $\pm$ 9.37	1.136	1.029–1.253	0.011
Octave confusion	4	8	8.921	1.719–46.307	0.009

Data are expressed as *n* (%) for categorical variables and as means $\pm$ SD for continuous variables. THI, tinnitus handicap inventory; diff., differences; PTA, pure tone audiometry.

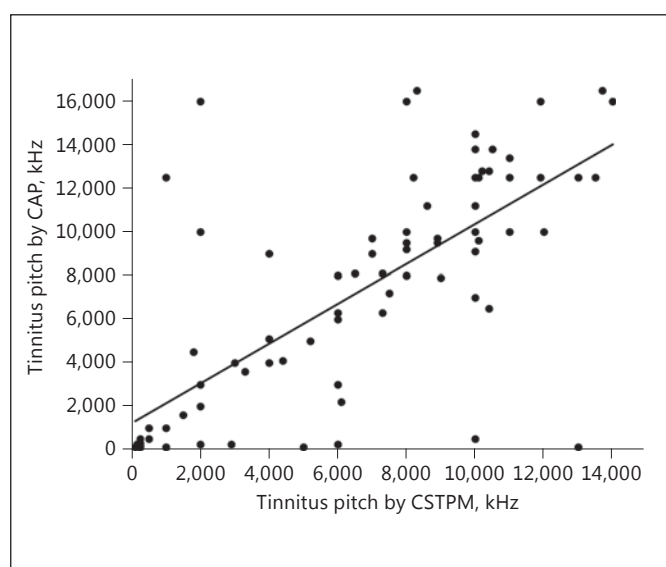


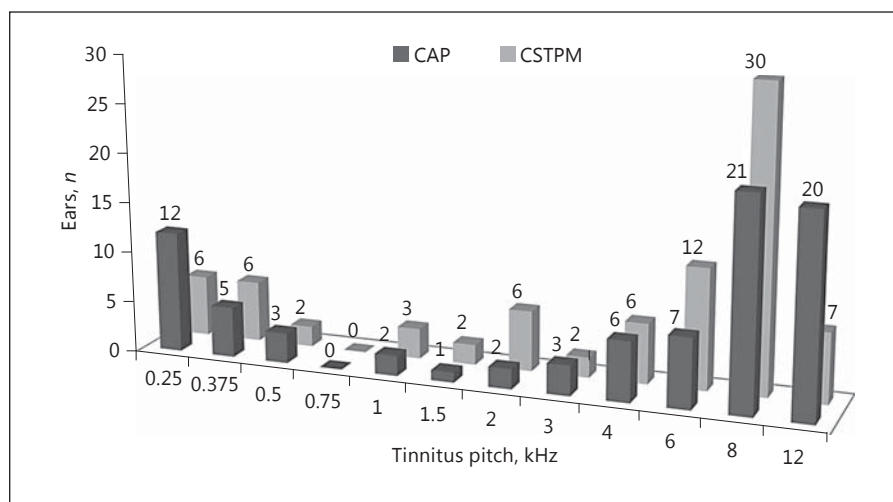
Fig. 2. Scatter plot showing the correlation of tinnitus pitch between the CAP and CSTPM. A significant positive correlation between the 2 procedure results was observed (Pearson $r = 0.793$, $p < 0.001$).

In total, 57 (69.5%) subjects matched their tinnitus pitch in the CSTPM with a difference ≤ 0.5 octaves from the CAP (Fig. 4). In the multivariate logistic regression, octave confusion showed the highest OR (8.92) of the tinnitus pitch differences between procedures. Tinnitus loudness was another significant contributing factor, but it increased the OR only slightly (OR 1.14; Table 2).

Discussion

Tinnitus assessment, including pitch matching, is traditionally performed as part of routine audiometric evaluations. The 2AFC method, the most common clinical testing protocol for tinnitus pitch matching, was described by Vernon and colleagues [Vernon and Meikle, 1981; Vernon and Fenwick, 1984; Vernon and Meikle, 1988] and involves the presentation of tones in pairs; the patient is asked to select the tone best matching the tinnitus pitch. Although the 2AFC procedure is the most common method used in the clinical setting, recent studies have suggested that 2AFC protocols have limitations. In Table 3, which summarizes past studies that compared conventional and automated tinnitus pitch-matching methods [Henry et al., 2004; Mahboubi et al., 2012; Wunderlich et al., 2015], 2 studies used 2AFC methods as a conventional or self-directed procedure [Henry et al., 2004; Wunderlich et al., 2015]. In both, mean pitch estimates differed significantly between 2AFC and non-2AFC methods. Henry et al. [2004] compared a 2AFC “manual” (conventional audiometric) procedure with 2 self-directed methods; 1 was 2AFC-based (“binary,” choosing between 2 frequencies separated by a third of an octave) and the other was non-2AFC-based (“subject-guided,” similar to the CSTPM in our study). In their study, pitch matches measured by the non-2AFC subject-guided method were significantly lower than those estimated by both 2AFC-based standard audiometric and binary self-administered methods. The authors argued that 2AFC-based algorithms imposed a restricted range of test

Fig. 3. Distribution of estimated tinnitus pitches by frequency for CAP and CSTPM.



frequencies and placed constraints on patient response choices, because response choices were always based on previous responses; this appeared to be responsible for the discrepancy between the procedures. Wunderlich et al. [2015] compared a 2AFC automated method with a conventional procedure, and found that the 2AFC automated method resulted in significantly lower pitch matches compared with the non-2AFC CAP. The authors suggested that because the first 2 frequency choices in their 2AFC method were always very distant from each other, errors in decisions usually resulted in a restricted frequency spectrum, which made returning to lower or higher frequencies for correcting the mistake impossible, and might have introduced a bias in the tinnitus pitch estimation.

An alternative method for tinnitus pitch matching consists of presenting a single initial tone (usually at 1,000 Hz), asking the patient if the tinnitus is higher or lower, and presenting the next tone an octave higher or lower depending on the answer. The procedure can continue in narrower steps depending on the ability of the patient to discriminate frequencies until the presented frequency matches the tinnitus [Pan et al., 2009; Mahboubi et al., 2012]. However, practical application of this method is often severely hindered by patients' inability to differentiate between "higher-" and "lower-pitch" concepts. Another known tinnitus pitch-matching method is the forced-choice double-staircase, FCDS, protocol [Penner and Bilger, 1992; Penner and Klafter, 1992]. Although this method has been reported to produce reliable tinnitus pitch matches, it requires a considerable time to complete and is thus rarely used in rapid clinical testing.

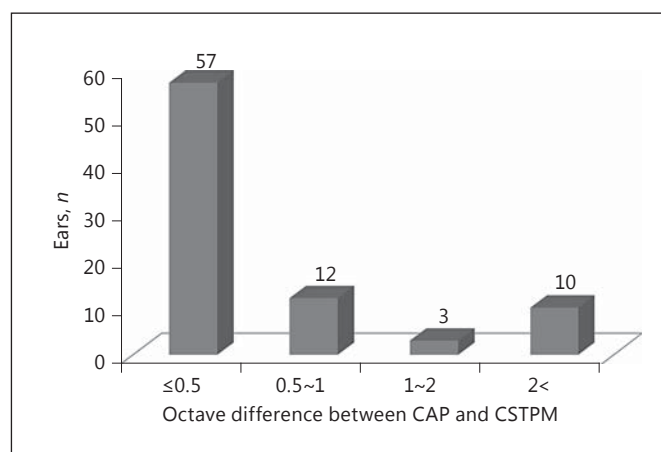


Fig. 4. Octave differences between the CAP and CSTPM tinnitus pitches showed that the tinnitus pitches measured by the 2 methods were very similar.

Given the practical difficulties associated with using these other methods in the clinical setting, the 2AFC protocol remains the most widely used method for tinnitus pitch matching due to its simplicity for patients and the relatively short time required to complete it. Because the purpose of the present study was to compare the self-directed tinnitus pitch-matching method with that commonly used during audiological evaluations, we used a common 2AFC-based procedure as the CAP (although we recognize the limitations of the 2AFC method). For the CSTPM, the slider method was chosen because it provides a very simple user interface and presents a great deal

Table 3. Summary of the studies that compared self-administered tinnitus pitch-matching methods with the conventional procedures

Study	Subjects, <i>n</i>	Equipment/test ear	Protocols	Mean tinnitus pitch	Results	Conclusion
Henry et al. [2004]	42	Laptop computer/ contralateral ear	2AFC (conventional) Binary-2AFC (self) Subject guided (self)	Conventional (S1: 4,762 Hz, S2: 6,334 Hz) Binary (S1: 6,436 Hz, S2: 6,643 Hz) Guided (S1: 4,942 Hz, S2: 5,438 Hz)	About 50% of subjects selected pitch matches over a range of 2 1/3 octaves	Specifying the range of pitch may be more appropriate
Mahboubi et al. [2012]	20	Desktop PC/ ipsilateral ear	Subject guided (conventional) Multiple choice (self) Slider (self)	Conventional (6,000 Hz) Multiple choice (6,000 Hz) Slider (5,925 Hz)	Self-matching is accurate as a standard protocol	Slider matching is more effective for individuals who might not find the exact match using the standard
Wunderlich et al. [2015]	17	iPod Touch/ contralateral ear	Subject guided (conventional) 2AFC (self)	Conventional (S1: 7,692 Hz, S2: 8,287 Hz) Self (S1: 6,126 Hz, S2: 6,140 Hz)	Test-retest reliability of the 2 methods was comparable	Self-matching is feasible without any loss of reliability or validity
Present study	82	Desktop PC/ contralateral ear	2AFC (conventional) Slider (self)	Conventional (6,983 Hz), Self (6,288 Hz)	Self-matching is appropriate as a conventional matching method	Octave confusion influences the difference in results between the procedures

2AFC, adaptive 2-alternative forced-choice procedure; S, session.

of flexibility and freedom for patients to choose the appropriate frequency without limiting their response to preselected choices. Additionally, the slider method has been shown to be preferred by patients experiencing a high tinnitus pitch [Mahboubi et al., 2012].

Another study compared 3 types of matching methods: conventional audiometric (presenting a single frequency and choosing the direction of presentation of the next frequency), multiple choice self-directed (choosing 1 frequency from among a series of frequency choices), and slider self-directed (the same as the CSTPM in our study) [Mahboubi et al., 2012] (Table 3). This is the only reported study that did not find significant differences between the pitch estimates of the different procedures. Only 20 subjects were involved in that study; our study had a considerably bigger sample size (82 subjects, the largest among the reported studies). We believe that the large sample size of our study elevates the reliability of its comparisons. No difference between the CAP and CSTPM results was found in our study. Additionally, there was a strong positive correlation between the CAP and CSTPM frequencies (Fig. 2). The pitch matches frequency distributions were similar

between the 2 methods (Fig. 3). Thus, the self-administered method used in our study seemed to estimate tinnitus pitch with comparable accuracy to that of the conventional procedure.

A major complication in tinnitus pitch matching is octave confusion. Octave confusion refers to the difficulty in distinguishing frequencies 1 octave apart from each other, and judging them as being identical. A common error made by patients during tinnitus pitch matching is selection of a sound that is 1 octave lower than the pitch of their tinnitus [Graham and Newby, 1962]. Confusion with a frequency 1 octave higher also occurs, albeit less frequently. Testing for octave confusion has been recommended as an intrinsic part of a regular tinnitus evaluation battery [Vernon et al., 1980]. All 3 studies on conventional versus self-directed tinnitus pitch method comparisons included testing for octave confusion in their pitch-matching procedures [Henry et al., 2004; Mahboubi et al., 2012; Wunderlich et al., 2015], but only 1 reported the octave confusion incidence: 25% in the study by Mahboubi et al. [2012]. None of these studies looked into the factors that might have influenced the difference in the results between the methods used. In

our study, only 14.5% of the patients experienced octave confusion, but octave confusion was found to be the factor that most influenced the difference in results between the 2 methods. Although choosing the proper frequency appears to be a simple task, research has shown that repeated pitch measurements produce inconsistent and highly variable results [Henry and Meikle, 2000]. The ability to match the tinnitus pitch to an external sound varies greatly from person to person. Musically gifted or trained individuals naturally perform better in frequency discrimination and pitch matching than “nonmusical” subjects [Beckett and Haggard, 1973; Spiegel and Watson, 1984; Kishon-Rabin et al., 2001; Tervaniemi et al., 2005; Bianchi et al., 2016]. For nonmusician patients, the acoustics-related concepts and terminology used during tinnitus pitch matching are often unfamiliar and confusing, which leads to errors such as octave confusion, or simply to wrong pitch estimates. Long-term frequency discrimination training is known to improve the frequency-distinguishing performance of nonmusicians [Menning et al., 2000]. However, long-term training may not be feasible for clinical application due to the time and staff requirements. Short-term unique training on a specific auditory task has been shown to be as effective as long-term training, and more beneficial than previous musical experience [Spiegel and Watson, 1984; Kishon-Rabin et al., 2001]. Thus, given the significant influence of octave confusion on the tinnitus pitch-matching results found in our study, we emphasize the importance of short-term tone discrimination training before performing tinnitus pitch matching. Incorporating such training into a regular tinnitus assessment routine is likely to improve the pitch-matching accuracy and reduce octave errors using either method.

We focused on investigating whether self-performed computer-based tinnitus pitch matching can be used instead of the routine procedure performed by an audiologist. The results of our study suggest that the computer-based automated method can provide measurement accuracy comparable to that of the most commonly used audiometric procedure, if the octave confusion issue can be adequately managed. The main advantages of the CSTPM over the conventional procedure are that it is simpler, easier, and quicker to complete. That the CAP is guided by an audiologist may appeal to an older population, for whom the computerized self-directed method may present a challenge due to their unfamiliarity with the technology and/or decreased dexterity and visual acuity. However, for the majority of the population, the self-directed method will not present any difficulties in execu-

tion, due to its simplicity and user friendly interface. The presence of an audiologist waiting for a response during the CAP may also put pressure on some patients and force errors. In contrast, performing the test on their own, and being able to change available frequencies back and forth at will and listen to them multiple times at their own pace, may give patients the time and freedom to produce more precise responses in a friendly environment. To obtain a reliable and precise pitch match, repeated measurements may be necessary because frequency discrimination is not a straightforward task for most people. The CSTPM lets subjects obtain as many pitch estimates as necessary without occupying clinical equipment or using audiometric test room time. Whereas the clinical procedure can only test for the frequencies available in the audiometer used, which can sometimes be limited, the CSTPM does not have limitations in terms of available frequencies, and suits patients with high-pitched tinnitus more than the CAP. Finally, being able to estimate tinnitus pitch independently, i.e., without using special hardware or assistance, is important for ensuring that patients can use the currently available software for tailor-made notched music training more consistently, which is an important condition for such therapies to be effective.

The CSTPM used in this study, including the user interface and tone-generating mechanism, is implemented as a part of the software package for tailored music therapy called Tinnitus Buster, which is freely available for download and use on PC-based Windows and Android-based mobile platforms (PC version: http://tinnitus.kangwon.ac.kr/?page_id=44; mobile version: <https://play.google.com/store/apps/details?id=com.tinnitus.musicfortinnitus.free>). The interface is currently offered only in Korean, but the update to English is planned once the program gets enough user support domestically.

It should be noted that this study only explored patients with tonal tinnitus. Therefore, our conclusions are limited to tonal tinnitus and may not be valid for generalization for other tinnitus types.

Conclusions

Considering the benefits of CSTPM, such as its relative time effectiveness, high number of measurements and frequency choices, and lower cost (due to not requiring clinical equipment or personnel), CSTPM can be an alternative method to CAP, provided that appropriate precautions are taken to prevent octave confusion.

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Disclosure Statement

The authors declare no conflicts of interest.

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